

Congruence Trace Universality and the Limits of First-Jet Separation in Bianchi Holonomy

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28 August 2026

Abstract

We examine whether a trace-divisibility statistic on a principal congruence subgroup of a Bianchi group can carry arithmetic information specific enough to initiate an orbit-owner construction. For a commutative ring R , a non-zero-divisor $m \in R$, and $\gamma = I + mA \in \mathrm{SL}_2(R)$, we prove the universal identity

$$\frac{\mathrm{tr}(\gamma)^2 - 4}{m^2} = m^2 \det(A)^2 - 4 \det(A).$$

Thus the integrality observed at Gaussian level 3 is a principal-congruence identity, not a Gaussian-specific signature. We also prove that the first congruence jet $J_m(\gamma) = A \bmod m$ is fixed by level- m conjugacy, changes sign under inversion, and is multiplied by the repetition number under powers. Exact computation on a frozen 11,481-matrix Gaussian word panel confirms every algebraic witness. The scalar discriminant takes only 145 values; adjoining the signed first jet raises the joint count to 517, separates 372 of 11,336 scalar-collision rows, and leaves 10,964 collisions, with the largest bucket reduced from 505 to 84 and no singleton joint buckets. Four frozen control families supply 6,396 exact matrices, but cover only two of three prespecified canonical control types. The result is consequently a rigorous negative-specificity theorem and a quantified necessary refinement, not an orbit-to-prime correspondence, a complete conjugacy classification, or a spectral realization. We state the remaining ownership and dynamical obligations explicitly.

Keywords: Bianchi group; principal congruence subgroup; holonomy trace; congruence jet; exact computation; arithmetic dynamics; negative control

Traditional Chinese Abstract / 繁體中文摘要

本文研究 Bianchi 群主同餘子群中的跡整除統計，能否提供足夠專屬的算術訊息以建立週期軌道所有者對應。對任意交換環 R 、非零因子 $m \in R$ ，以及 $\gamma = I + mA \in \mathrm{SL}_2(R)$ ，我們證明標準化跡判別式恆等式 $(\mathrm{tr}(\gamma)^2 - 4)/m^2 = m^2 \det(A)^2 - 4 \det(A)$ 。因此，高斯整數層級三所見的整性並非高斯算術的專屬特徵，而是普遍的主同餘代數結果。本文另證明第一同餘噴射 $A \bmod m$ 在層級共軛下不變、取逆時變號、取冪時依重複次數縮放。於凍結的一萬一千四百八十一個高斯矩陣精確面板中，所有代數證書均通過；標量判別式僅有一百四十五類，加入帶符號第一噴射後增至五百一十七類，雖分離部分碰撞，仍留下大量不可區分矩陣。四組控制族亦證實公式跨整數、高斯鄰近層級及 Eisenstein 整數成立。故本文成果是負向專屬性定理與必要描述子的量化改良，而非素理想所有者映射、完整共軛分類或譜實現；尚缺的控制類型、軌道完備性與動力學橋接均被明確保留為開放問題。 關鍵詞：Bianchi 群；主同餘子群；全純跡；同餘噴射；精確計算；算術動力學；負向控制

1 Introduction

Closed geodesics on a finite-volume hyperbolic three-manifold are represented by loxodromic conjugacy classes, and their complex lengths are encoded by holonomy eigenvalues and traces. This makes trace expressions attractive candidates for connecting dynamics to arithmetic. It also creates a danger: a congruence identity can look arithmetically distinctive inside one ring while actually being forced by determinant one over every commutative base ring. The distinction matters before any stronger orbit-owner, Euler-product, or spectral claim is attempted.

Arithmetic Kleinian groups and their finite-index congruence subgroups provide a standard geometric setting for this question [3, Chs. 10–11]. Congruence subgroups of Bianchi groups support substantial analytic and topological structure, including torsion questions for coefficient systems [5, Introduction and Sec. 2]. On finite-volume hyperbolic manifolds, dynamical zeta functions organize closed-geodesic data, but the analytic object requires the full geometric and representation-theoretic setting rather than a finite word sample [4, abstract and Secs. 3–5]. These facts motivate a narrower question:

Does normalized trace divisibility at Gaussian level 3, perhaps refined by the first congruence jet, distinguish candidate holonomy owners in a way that is specific to the Bianchi arithmetic source?

Our answer has two parts. First, normalized trace divisibility is universal. If $\gamma = I + mA \in \mathrm{SL}_2(R)$, determinant one forces $\mathrm{tr}(A) = -m \det(A)$, and hence

$$D_{m^2}(\gamma) := \frac{\mathrm{tr}(\gamma)^2 - 4}{m^2} = m^2 \det(A)^2 - 4 \det(A).$$

No unique factorization, imaginary quadratic structure, or hyperbolic geometry is used. This refutes the specificity of D_9 as a Gaussian owner mechanism while retaining it as an exact necessary invariant.

Second, the first jet $J_m(\gamma) = A \bmod m$ carries information that scalar trace loses. We prove its exact level-conjugacy, inversion, and power laws. On the frozen Gaussian panel, the joint descriptor $(D_9, [J_3]_{\pm})$ substantially reduces the worst scalar collision bucket but remains far from injective. The finite audit has no target prime or zero labels and is not used to tune a statistic. It is a reproducible stress test of already stated algebraic laws.

The contribution is deliberately asymmetric. The positive result is a general theorem and a lawful first-jet refinement. The model-selection result is negative: the scalar statistic proves too much, and the refinement still separates too little. We do not infer an orbit-to-prime-ideal map, identify rational primes, fit zeta zeros, claim a complete primitive-orbit census, or invoke a fallback spectral route. This claim boundary is part of the result.

2 Geometric and algebraic setting

2.1 Bianchi holonomy and the frozen flow

Let $K = \mathbb{Q}(i)$, let $\mathcal{O}_K = \mathbb{Z}[i]$, and write

$$\Gamma(3) = \ker(\mathrm{SL}_2(\mathbb{Z}[i]) \longrightarrow \mathrm{SL}_2(\mathbb{Z}[i]/(3))).$$

After projectivization, a torsion-free finite-index subgroup acts freely on hyperbolic three-space, and unit-speed geodesic flow uses arclength as its clock. The frozen object is the geodesic flow on the level-(3) Bianchi quotient, with primitive loxodromic conjugacy classes as primitive owners and group powers as

repetitions. The computation samples an elementary-generator word ball inside this group; it does not enumerate all conjugacy classes or every element in a geometric-length range.

For a loxodromic element, the eigenvalue pair determines translation length and rotational holonomy, so trace data naturally participate in the complex length spectrum [2, Introduction and Sec. 2]. Trace is invariant under conjugacy and inversion in SL_2 , but it need not be a complete conjugacy invariant.

Lemma 2.1 (level-three torsion exclusion). *If $\gamma \in \Gamma(3) \subset \mathrm{SL}_2(\mathbb{Z}[i])$ has finite order, then its image in $\mathrm{PSL}_2(\mathbb{C})$ is the identity.*

Proof. Write $\gamma = I + 3A$. Determinant one gives $\mathrm{tr}(\gamma) = 2 - 9\det(A)$, hence $\mathrm{tr}(\gamma) \equiv 2 \pmod{9}$. A finite-order matrix in $\mathrm{SL}_2(\mathbb{C})$ is diagonalizable with roots of unity λ, λ^{-1} , so its trace is a real algebraic integer in $\mathbb{Z}[i]$, hence an integer. The possible integral traces are $-2, -1, 0, 1, 2$. Only 2 is congruent to 2 modulo 9; therefore both eigenvalues equal 1, and finite order plus diagonalizability gives $\gamma = I$. The central element $-I$ is excluded by the congruence. $\square$

2.2 Evidence types and owner compatibility

We separate three logical levels:

- (i) a ring identity proved for every admissible matrix;
- (ii) an exact finite computation verifying implementations and measuring collisions on a frozen panel;
- (iii) a dynamical or arithmetic ownership claim, which additionally requires a complete primitive-owner rule and source-derived labels.

Only the first two are established. An invariant under conjugacy and inversion is merely eligible to be attached to an unoriented orbit. Eligibility is not ownership: a useful owner coordinate must control collisions, respect repetition, and relate to an independently specified arithmetic source.

Definition 2.2 (normalized discriminant and first jet). Let R be a commutative ring and let $m \in R$ be a non-zero-divisor. For $\gamma = I + mA \in \mathrm{SL}_2(R)$, the matrix A is unique; define

$$D_{m^2}(\gamma) = \frac{\mathrm{tr}(\gamma)^2 - 4}{m^2}, \quad J_m(\gamma) = A \bmod m \in M_2(R/mR).$$

For unoriented owners use the signed class $[J_m]_{\pm} = \{J_m, -J_m\}$.

The notation D_{m^2} records the normalizing ideal power and prevents D_9 from being mistaken for a construction intrinsic to $\mathbb{Z}[i]$.

3 Related work and methodological controls

Selberg-type zeta functions on odd-dimensional finite-volume hyperbolic manifolds admit a rigorous analytic theory involving regularized traces, scattering terms, and representation data [4, abstract and Secs. 3–5]. Nothing in the finite word-ball computation supplies that apparatus. Our statistic is a candidate algebraic owner coordinate, not a zeta function or spectral determinant.

Modern work on Bianchi congruence towers illustrates how arithmetic and topology can interact at a source-owned level [5, Introduction and Sec. 2]. It motivates congruence invariants but does not imply that trace divisibility at one level is Gaussian-specific.

For geometric controls, verified computation matters when a numerical manifold presentation is used. Interval methods can certify a complete hyperbolic structure for a cusped triangulation [1, Thm. 5.1 and

Secs. 3–5]. Official documentation distinguishes verified interval results from high-precision calculations [7, `verified computations` and `verify_hyperbolicity`]. Earlier development used the 5_2 knot complement as a finite-volume, one-cusped, non-arithmetic comparison. Its non-arithmetic role follows from the arithmetic knot-complement classification [6, main theorem]. The earlier numerical length prefix is contextual only and is not an input here.

Our control is stricter: derive the identity before reading target labels, replay it over rings and levels that remove Gaussian specificity, and report failure of injectivity. The conclusion does not depend on an asymptotic approximation or visual match.

3.1 Position of the present result

The manuscript separates three questions that can otherwise be conflated. The first is whether a formula is well-defined and invariant. The second is whether its values distinguish primitive dynamical owners. The third is whether the distinguished owners carry the source arithmetic required by a proposed Euler product. The universal trace theorem completely answers the first question for the frozen scalar. The first-jet theorem improves the answer to the second question but the collision ledger shows that it does not complete it. The third question remains open because no prime-ideal label is attached independently of the descriptor.

This ordering is important. If arithmetic labels are consulted before the invariant and its null controls are frozen, an expressive finite descriptor can be adjusted until it appears to track almost any target table. Here the statistic is derived from determinant one, its power rule is proved symbolically, and its cross-ring behavior is predicted before the control replay. The negative specificity verdict therefore does not depend on a fitted threshold, selected subset, or favorable plotting scale.

The result also differs from a claim that trace has no geometric value. Trace determines eigenvalues for a determinant-one 2×2 matrix and therefore carries complex-length information for loxodromic holonomy. The failure is narrower: the particular normalized divisibility of trace is not a Gaussian-specific arithmetic owner, and first-order congruence data do not separate the frozen matrices. A future construction can still use trace as one coordinate, provided additional source-owned data and equivalence rules are supplied.

Finally, the paper distinguishes a necessary invariant from a sufficient code. Conjugacy invariance is necessary if an unoriented closed geodesic is the owner. Inversion compatibility is necessary because reversing orientation should not silently create a new scalar label under the frozen convention. A power law is necessary to distinguish a primitive owner from its traversals. None of these properties alone gives injectivity, primitivity, or arithmetic ownership. The collision experiment measures this remaining gap.

4 Universal trace and jet theorems

Lemma 4.1 (determinant expansion). *For every commutative ring R , scalar $m \in R$, and $A \in M_2(R)$,*

$$\det(I + mA) = 1 + m \operatorname{tr}(A) + m^2 \det(A).$$

Proof. Writing $A = (a_{jk})$, direct expansion gives

$$(1 + ma_{11})(1 + ma_{22}) - m^2 a_{12}a_{21} = 1 + m(a_{11} + a_{22}) + m^2(a_{11}a_{22} - a_{12}a_{21}).$$

□

Theorem 4.2 (congruence trace universality). *Let R be a commutative ring, let $m \in R$ be a non-zero-divisor, and let $\gamma = I + mA \in \mathrm{SL}_2(R)$. Then*

$$\mathrm{tr}(A) = -m \det(A), \quad \mathrm{tr}(\gamma) = 2 - m^2 \det(A),$$

and

$$D_{m^2}(\gamma) = \frac{\mathrm{tr}(\gamma)^2 - 4}{m^2} = m^2 \det(A)^2 - 4 \det(A) \in R. \quad (1)$$

Proof. By lemma 4.1 and $\det(\gamma) = 1$,

$$m \mathrm{tr}(A) + m^2 \det(A) = 0.$$

Because m is a non-zero-divisor, cancellation yields $\mathrm{tr}(A) = -m \det(A)$. Thus $\mathrm{tr}(\gamma) = 2 - m^2 \det(A)$, and

$$\mathrm{tr}(\gamma)^2 - 4 = -4m^2 \det(A) + m^4 \det(A)^2 = m^2(m^2 \det(A)^2 - 4 \det(A)).$$

The quotient is an identity in R , not informal division in a fraction field. $\square$

Corollary 4.3 (Gaussian level three). *For every $\gamma = I + 3A \in \mathrm{SL}_2(\mathbb{Z}[i])$,*

$$D_9(\gamma) = \frac{\mathrm{tr}(\gamma)^2 - 4}{9} = 9 \det(A)^2 - 4 \det(A) \in \mathbb{Z}[i].$$

The same formula holds over $\mathbb{Z}$, $\mathbb{Z}[\omega]$, and every commutative ring in which the level is a non-zero-divisor.

Gaussian D_9 -integrality is true and useful as a checksum, yet it cannot diagnose Gaussian arithmetic. It persists after the supposed source is replaced.

4.1 Additional consequences of universality

The proof of theorem 4.2 yields useful diagnostic consequences. First, the normalized discriminant factors through the single scalar $\det(A)$. At fixed m , every pair A, A' with $\det(A) = \det(A')$ has the same D_{m^2} , irrespective of their other effective matrix coordinates. Large collisions are therefore structural rather than an implementation defect. The 145-value result in the word panel reflects this forced compression.

Second, the non-zero-divisor hypothesis marks the cancellation boundary. The expansion $m(\mathrm{tr}(A) + m \det(A)) = 0$ holds without it, but $\mathrm{tr}(A) = -m \det(A)$ need not follow if m has a nontrivial annihilator. The theorem is not stated for arbitrary zero-divisor levels, and the code does not use such a case as a control. Over $\mathbb{Z}$, $\mathbb{Z}[i]$, and $\mathbb{Z}[\omega]$, every nonzero level is a non-zero-divisor because these rings are domains.

Third, for a fixed trace t , proposition 4.5 shows that repetitions move along a prescribed square-factor sequence $S_{r-1}(t)^2$. This is an exact consistency test for a claimed traversal. Its converse is false: observing a compatible square factor does not prove that one matrix is a power of another. The finite audit uses the identity only in the forward direction.

The first jet resolves information discarded by determinant. In dimension two, $\det(A)$ is one polynomial coordinate while $A \bmod m$ retains all four residue entries subject to a first-order trace condition. This explains why the joint count grows. It also explains why first order cannot be complete: infinitely many level matrices have the same residue, and ambient conjugacy can move residues within an adjoint orbit. The signed convention handles inversion only; it does not quotient the full ambient adjoint action.

Proposition 4.4 (compatibility of scalar and jet traces). *Under the hypotheses of theorem 4.2, the first jet satisfies*

$$\mathrm{tr}(J_m(\gamma)) = 0 \quad \text{in } R/mR.$$

Proof. The identity $\text{tr}(A) = -m \det(A)$ reduces to $\text{tr}(A) \equiv 0 \pmod{m}$. Since $J_m(\gamma) = A \pmod{m}$, its trace vanishes in the quotient. $\square$

Thus the jet lies in the trace-zero tangent space at the identity. This makes the term first jet literal: it is the first-order displacement inside the determinant-one group scheme. The proposition is still universal and earns no Gaussian specificity.

Proposition 4.5 (trace recurrence and powers). *Let $\gamma \in \text{SL}_2(R)$, $t = \text{tr}(\gamma)$, and define*

$$S_0(t) = 1, \quad S_1(t) = t, \quad S_n(t) = tS_{n-1}(t) - S_{n-2}(t) \quad (n \geq 2).$$

For $r \geq 1$,

$$\text{tr}(\gamma^r)^2 - 4 = (t^2 - 4)S_{r-1}(t)^2.$$

Consequently $D_{m^2}(\gamma^r) = D_{m^2}(\gamma)S_{r-1}(t)^2$.

Proof. Cayley–Hamilton gives $\gamma^2 - t\gamma + I = 0$. Induction yields $\gamma^r = S_{r-1}(t)\gamma - S_{r-2}(t)I$, and then $\gamma^r - \gamma^{-r} = S_{r-1}(t)(\gamma - \gamma^{-1})$. Taking eigenvalues in the universal quadratic extension, or comparing determinants over $\mathbb{Z}[t]$, proves the identity. Both sides are integral polynomials in t , so it specializes to every commutative ring. $\square$

Conjugacy and inversion invariance follow from trace invariance and $\text{tr}(\gamma^{-1}) = \text{tr}(\gamma)$. The square factor is repetition compatible but not a primitive-root test.

Theorem 4.6 (first congruence jet laws). *Let R be a commutative ring, let $m \in R$ be a non-zero-divisor, set $\Gamma(m) = \{I + mB \in \text{SL}_2(R)\}$, and let $\gamma = I + mA \in \Gamma(m)$. Then:*

- (a) $J_m(h\gamma h^{-1}) = J_m(\gamma)$ for every $h \in \Gamma(m)$;
- (b) $J_m(\gamma^{-1}) = -J_m(\gamma)$;
- (c) $J_m(\gamma^r) = rJ_m(\gamma)$ for every $r \geq 1$, with r read in R/mR .

Thus $[J_m]_{\pm}$ descends to an unoriented level-conjugacy descriptor.

Proof. Modulo m^2 ,

$$(I + mB)(I + mA)(I + mB)^{-1} \equiv I + mA,$$

because $(I + mB)^{-1} \equiv I - mB$. This proves (a). Similarly $(I + mA)^{-1} \equiv I - mA$, proving (b). The binomial expansion gives $(I + mA)^r \equiv I + r mA \pmod{m^2}$, proving (c). $\square$

Remark 4.7. The invariance is under $\Gamma(m)$ -conjugacy. A larger ambient group acts on the jet by the reduced adjoint action. We do not call J_m a complete invariant of ambient Bianchi conjugacy.

4.2 Constructive non-injectivity

Take

$$A_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix}, \quad \gamma_j = I + 3A_j.$$

Both determinants vanish, so $D_9(\gamma_1) = D_9(\gamma_2) = 0$. Their jets are distinct up to sign because $i \not\equiv \pm 1 \pmod{3}$, so the jet strictly refines the scalar. Conversely, upper-unitriangular matrices with the same upper-right residue have the same joint descriptor; the joint descriptor is not injective.

5 Exact finite certificate

5.1 Frozen panel and arithmetic representation

The computation starts from a prespecified elementary-generator word ball through reduced word length five. It contains 22,409 reduced words and canonicalizes them to 11,481 exact 2×2 Gaussian matrices. Gaussian integers are ordered pairs of machine integers, and all ring operations are exact. No floating-point tolerance enters the Round-7 or Round-8 certificate.

For each matrix, the program checks determinant one and congruence to I modulo 3, constructs $A = (\gamma - I)/3$, evaluates both sides of [equation \(1\)](#), and checks the power identity for $r = 1, \dots, 5$. It canonicalizes the signed jet by taking the lexicographically smaller residue encoding of J_3 and $-J_3$.

The panel is exhaustive only relative to its word-generation and cutoff contract. Different words can yield the same matrix, and elements outside the ball may be conjugate to elements inside it. The rows are not a complete primitive-orbit ledger.

5.2 Certificate algorithm

The certificate is a sequence of independently checkable transformations. First, a reduced word is evaluated in the frozen elementary generators. Second, exact matrix entries are canonicalized as Gaussian pairs and duplicate matrices are removed. Third, the program asserts determinant one and verifies that every entry of $\gamma - I$ is divisible by 3. Fourth, it constructs A by exact coordinate division, never floating-point rounding. Fifth, it evaluates trace, determinant, both sides of [equation \(1\)](#), and recurrence witnesses for five repetitions. Finally, it reduces A modulo 3, canonicalizes the signed jet, and builds scalar and joint bucket dictionaries.

Failure modes are explicit. A non-divisible entry stops the row rather than coercing a quotient. A determinant mismatch is reported before descriptor construction. A formula disagreement fails verification. Collision counts are recomputed from keys rather than read from a summary. The verify-only path reads the frozen ledger and receipt but does not rewrite them, so reproduction cannot silently replace the evidence it checks.

For buckets of sizes $b_1, \dots, b_k$, the reported collision-row quantity is

$$\sum_{j=1}^k (b_j - 1) = N - k,$$

where N is panel size and k the number of descriptors. Thus $11,481 - 145 = 11,336$ scalar collision rows and $11,481 - 517 = 10,964$ joint collision rows. Their difference 372 is exactly the number resolved by the jet. This avoids pair-count inflation: it measures rows beyond first representatives, not all $\binom{b_j}{2}$ colliding pairs.

The maximum bucket probes a different failure mode from total collisions. Reducing it from 505 to 84 shows that the jet prevents one scalar value from dominating as severely. The absence of singleton joint buckets shows that no row is uniquely identified even at finite matrix level. Reporting both prevents a favorable average from hiding a worst case or a favorable worst case from hiding pervasive collisions.

The jet increases descriptor count by about 3.57 and reduces the maximum bucket by about 6.01. These are meaningful finite gains but not owner separation: more than 95% of rows remain collisions beyond first occurrences, and no joint bucket is a singleton. Matrix equality is finer than conjugacy ownership, so these counts cannot be promoted to distinct primitive geodesics.

Table 1: Exact collision profile on the frozen Gaussian panel.

Quantity	Exact count
Matrices	11,481
Distinct scalar D_9 values	145
Scalar collision rows	11,336
Distinct joint $(D_9, [J_3]_{\pm})$ descriptors	517
Scalar-collision rows separated by the jet	372
Joint collision rows remaining	10,964
Largest scalar bucket	505
Largest joint bucket	84
Singleton joint buckets	0

Table 2: Frozen controls for the universal identity.

Family	Level	Rows	Result
Full ambient $\mathrm{SL}_2(\mathbb{Z}[i])$ witnesses	—	4	3 nonintegral, 1 integral
Integer principal congruence	3	485	all pass
Gaussian neighboring levels	2 and 4	1,969 each	all pass
Eisenstein principal congruence	3	1,969	all pass

5.3 Cross-ring and neighboring-level controls

The theorem was replayed on four frozen families containing 6,396 exact matrices or witnesses. The principal-congruence subset has 6,392 rows, all passing. Four extra full-ambient Gaussian matrices test the boundary: three produce a nonintegral quotient by 9, while one happens to be integral.

The controls cover a simpler parent/source replacement and neighboring parameters. Under the prespecified Route-A evaluator, they instantiate only two of three required canonical control types. Four executed families are not reported as a complete three-type gate. The algebraic refutation is complete, while the broader protocol retains an open control obligation.

5.4 Reproducibility contract

The repository includes generating programs, all historical unit tests, frozen JSON contracts, CSV ledgers, validation notes, and a verify-only Round-8 reproducer. Verification reads committed artifacts, checks schema and hashes, regenerates exact core results in memory, and compares them without overwriting evidence. No prime table, zeta-zero table, or tuned threshold is used.

The certificate establishes agreement with theorems and finite counts. It cannot establish full-group generation at a geometric cutoff, complete conjugacy, or an arithmetic label map.

5.5 Why the finite panel remains informative

An incomplete word ball cannot prove an asymptotic owner theorem, but it can decisively test exact implementation claims and expose non-injectivity. One counterexample refutes injectivity, while thousands of collisions quantify its scale under a frozen enumeration. One source-removed family can motivate skepticism about specificity; the universal proof upgrades that skepticism to a theorem.

The panel is reusable. If a higher jet or ideal-valued coordinate is proposed, it can be evaluated on the same 11,481 matrices without changing enumeration, permitting a paired comparison of counts and buckets. A successor should be frozen before its collision profile is read and should add a genuinely third

canonical control type. Reusing the panel is legitimate for controlled comparison; repeatedly optimizing a coordinate against observed buckets would be exploratory feature engineering and must be labeled accordingly.

6 Adversarial interpretation and Route-A assessment

6.1 What the controls refute

An arithmetic-specific owner statistic should deteriorate when its source is removed or replaced. Here the normalized discriminant survives over $\mathbb{Z}$, neighboring Gaussian levels, and $\mathbb{Z}[\omega]$ for the same elementary reason. The explanation is [theorem 4.2](#), not hidden Gaussian prime-ideal structure. The statistic is stopped as a Gaussian-specific mechanism.

This is an identity-level refutation, not histogram similarity. More computation at level three cannot restore specificity. Progress requires an independently source-derived coordinate, such as an ideal- or Hecke-valued refinement, whose laws are proved and whose content fails on source-removed controls.

6.2 Typed Route-A boundary

The proxy is the marked-word congruence descriptor, not the complete Bianchi flow. Its typed assessment is

$$(A0_{\text{weak arithmetic relation}}, A1_{\text{weak}}, A2_{\text{fail}}, A3_{\text{fail}}, A4_{\text{fail}}),$$

with exploratory overall status.

- **A0:** the ambient group is arithmetic, but no independent ideal owner is attached to a row; the formula is not source-specific.
- **A1:** transformation laws are exact, but the census is incomplete and the descriptor has heavy collisions.
- **A2:** no dynamical trace formula or determinant is derived.
- **A3:** no arithmetic Euler factor is reconstructed.
- **A4:** no spectral target is identified.

The full Bianchi flow remains unassigned because proxy credit cannot cross an ownership gap. No alternate spectral construction is activated: no self-adjoint operator, boundary condition, spectral reality proof, or zero correspondence is supplied.

6.3 Alternative explanations and what would change the verdict

One explanation for repeated D_9 values is that many words encode the same conjugacy class. Duplicate matrices are already removed, so word equality alone cannot explain the collisions. Unknown conjugacies could merge remaining matrices into fewer owners, but that would not make the descriptor more injective at matrix level and cannot restore Gaussian specificity against [theorem 4.2](#).

Another possibility is that only a selected primitive loxodromic subset matters. Restricting the panel could reduce collisions, but selection must come from an independently proved primitive and geometric ownership rule. Choosing rows after seeing descriptor values would not establish ownership. Moreover, cross-ring universality remains: any specificity of a restricted scalar would arise from the selection rule, not from D_9 integrality.

A result strong enough to change the verdict would contain at least four components. It would define an ideal-valued label from the Gaussian source without target fitting; prove conjugacy, inversion, and repetition laws; show exact or certified separation on a complete declared owner domain; and fail prespecified source-removed controls for a reason linked to Gaussian splitting. Only then would it be meaningful to ask whether a convergent dynamical product reconstructs independently specified local factors.

6.4 Falsifiers for a successor statistic

A successor should be rejected if it is determined by the universal determinant-one expansion; changes under declared owner equivalence; lacks a primitive/repetition law; remains equally structured after integer or Eisenstein replacement; or requires reading target primes or zeros during design. Passing these tests is still only a prerequisite. A complete proposal needs auditable primitive ownership, convergence or regularization, and an explicit source-label-to-Euler-factor route.

6.5 Implications for flow modeling

The frozen continuous-time object is more structured than the matrix panel. A primitive loxodromic class carries a geometric length and rotational angle, while the word panel is organized by presentation length. Any flow product must use the geometric clock and must avoid counting the same unoriented conjugacy class several times through cyclic shifts, inversion, or alternative words. The present descriptor laws are compatible with some of these identifications, but the artifact does not certify the complete quotient by all of them.

This difference explains why the negative theorem can be transferred upward while positive separation credit cannot. Universality is proved for every admissible matrix, so it applies in particular to all holonomies that arise from the flow: D_9 remains nonspecific. By contrast, a favorable finite collision rate would concern only the sampled matrices and could not prove complete flow ownership. Negative identity-level information is monotone under restriction; finite injectivity evidence is not monotone under expansion to an unseen group.

The same logic governs repetitions. The recurrence in [proposition 4.5](#) verifies how a known power transforms. It does not decide whether a word is primitive in the full group, because a shorter root may lie outside the cutoff or use a different presentation. A flow-facing owner ledger therefore needs an independent primitive-root procedure with a completeness statement, not merely a within-ball word test.

Finally, a future dynamical product would need analytic control that no finite collision ledger provides. One must specify its convergence half-plane or regularization, multiplicities, treatment of cusps and scattering, and the exact role of rotational holonomy. The source literature shows that finite-volume zeta theory contains these ingredients, but it does not turn the current congruence statistic into them. The paper stops at the strongest conclusion justified by its algebraic and certificate layers.

7 Limitations and open obligations

First, the finite ledger is an elementary-subgroup word ball, not a theorem about all of $\Gamma(3)$. Word length is presentation dependent and is not hyperbolic length. Full-group generation, conjugacy, and primitivity remain open.

Second, J_3 is invariant under level-subgroup conjugacy, whereas ambient conjugacy acts nontrivially. A future descriptor may need an adjoint-orbit class. Heavy collisions show first-order data are insufficient. Higher jets might refine the panel, but their specificity needs new controls rather than larger tables.

Third, controls do not match every geometric feature. Earlier comparisons improved finite-volume and cusp matching through the 5_2 complement, but cusp count, covolume, length distribution, and marked presentation are not simultaneously matched. These mismatches do not affect the universal refutation, but matter for new dynamical statistics.

Fourth, a $\mathbb{Q}(i)$ Dedekind-zeta calibration remains only a modeling direction. It requires a source-owned prime-ideal rule, including norm multiplicities and split, inert, and ramified behavior under rational-prime push-forward. No such map appears here. Nothing provides evidence for Riemann-zeta zeros.

Finally, this is not a nonexistence theorem for all arithmetic encodings of Bianchi geodesics. It rules out the frozen scalar mechanism and quantifies one first-order refinement. Ideal-valued holonomy data, Hecke correspondences, and geometric transfer operators remain outside the tested hypothesis.

8 Conclusion

Normalized trace divisibility at Gaussian level three is exact but universal: determinant one forces the same identity in every commutative principal-congruence setting with a cancellable level. The first jet has clean owner-compatible laws and improves finite separation, yet leaves 10,964 of 11,336 scalar-collision rows unresolved and no singleton joint buckets. Exact controls stop D_9 as a Gaussian-specific owner mechanism while preserving the theorem, software, and ledger as reusable infrastructure. The next meaningful step is a source-derived ideal-valued refinement, a third canonical control type, and a complete ownership audit.

Declarations

Funding

The author received no funding for this work.

Conflict of Interest

The author declares no conflicts of interest.

Author contributions

Liang Wang: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing—original draft; Writing—review and editing; Visualization; Project administration.

Data and code availability

All frozen inputs, exact ledgers, validation records, source code, unit tests, and verify-only reproduction scripts supporting this study are in the accompanying repository under `papers/24-bianchi-holonomy-flow/`. No restricted, personal, or proprietary dataset was used. Artifact hashes and commands are recorded in `paper/stage2_manuscript_audit.md`.

Ethics statement

This theoretical and computational study involved no human participants, personal data, animals, clinical interventions, or biological materials. Institutional ethics approval and informed consent were not applicable.

AI-Assisted Research Disclosure

Generative-AI tools assisted language drafting, LaTeX preparation, code-oriented consistency checks, and manuscript-structure review. The author directed the research, fixed the claim boundary, inspected derivations and artifacts, verified cited sources, and accepts full responsibility for accuracy and integrity. No AI system is listed as an author.

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